

House Price Prediction

Using tabular data & satellite imagery

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PROBLEM STATEMENT

The objective of this project is to develop a **multimodal regression pipeline** that accurately predicts residential property prices by jointly leveraging structured tabular data and unstructured satellite imagery. Traditional valuation models rely primarily on numerical attributes such as size, location, and construction quality, but often fail to capture environmental and neighborhood-level factors that influence market value.

Given historical housing data with geographic coordinates, the task is to programmatically acquire satellite images for each property, extract meaningful visual features using CNNs, and fuse them with tabular features to improve price prediction. The model must be interpretable, enabling analysis of both tabular feature importance and visual regions influencing predictions through techniques such as Grad-CAM.

APPROACH OVERVIEW

The project follows a staged multimodal learning pipeline. First, exploratory data analysis and preprocessing were performed on the provided tabular housing dataset to understand feature distributions, correlations, and target behavior. The target variable (price) was log-transformed to address skewness and stabilize variance.

Next, satellite imagery was programmatically acquired for each property using latitude and longitude coordinates via the Mapbox Static Images API. These images capture neighborhood-level environmental context unavailable in tabular form. A pretrained ResNet-18 model was used as a frozen feature extractor to convert each satellite image into a 512-dimensional visual embedding. To reduce dimensionality and noise, Principal Component Analysis (PCA) was applied to the image embeddings.

Three modeling strategies were then evaluated using XGBoost regression: a baseline model using only tabular features, an image-only model using PCA-reduced image embeddings, and a hybrid model formed by concatenating scaled tabular features with PCA-reduced image features. Model performance was compared using RMSE and R^2 in log-price space.

Finally, model interpretability was addressed through feature importance analysis for tree-based models and Grad-CAM visualizations on satellite images, enabling inspection of both tabular feature contributions and spatial regions influencing visual representations.

PIPELINE

- Load and preprocess tabular housing data
 - Handle feature selection and scaling
 - Apply log transformation to the target price
- Programmatically fetch satellite images
 - Use latitude and longitude with Mapbox Static Images API
 - Store images in train and test directories
- Extract visual features from images
 - Use pretrained ResNet-18 as a frozen feature extractor
 - Generate 512-dimensional embeddings per image
- Reduce image feature dimensionality

- Apply PCA to image embeddings
 - Retain principal components capturing most variance
- Construct modeling datasets
 - Tabular-only dataset
 - Image-only dataset (PCA-reduced embeddings)
 - Hybrid dataset (concatenated tabular + image features)
- Train and evaluate models
 - XGBoost on tabular data (baseline)
 - XGBoost on image features
 - XGBoost on fused hybrid features
- Explain model behavior
 - Feature importance from XGBoost
 - Grad-CAM visualizations on satellite images

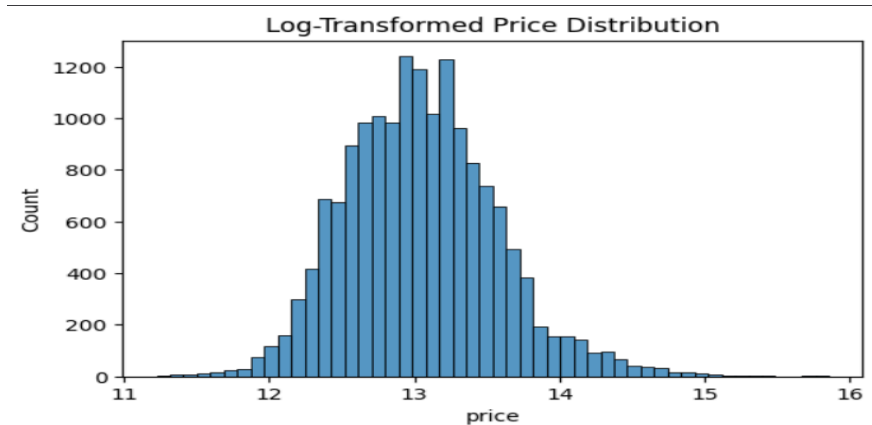
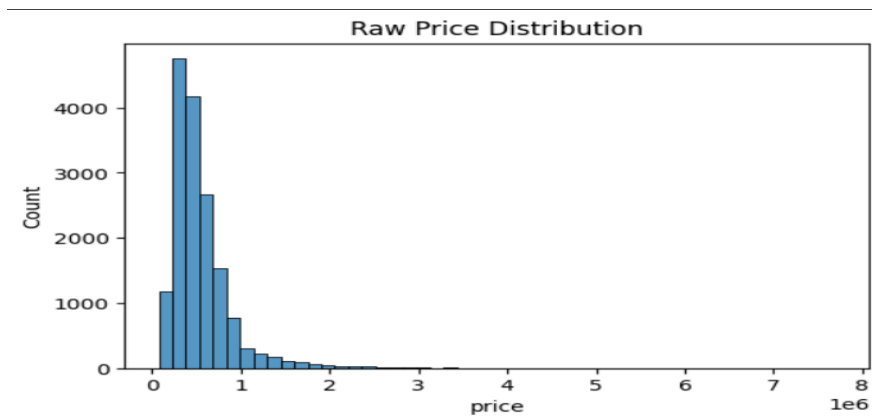
RESULTS

- **Baseline (Tabular only):**
 - RMSE (log): **0.1689**
 - R^2 : **0.8966**
- **Image-only (CNN embeddings + PCA):**
 - RMSE (log): **0.4314**
 - R^2 : **0.3256**
- **Hybrid (Tabular + Image PCA):**
 - RMSE (log): **0.1660**
 - R^2 : **0.9001**
- **Improvement (Hybrid vs Baseline):**
 - RMSE ↓ **0.0029**
 - R^2 ↑ **0.0035**

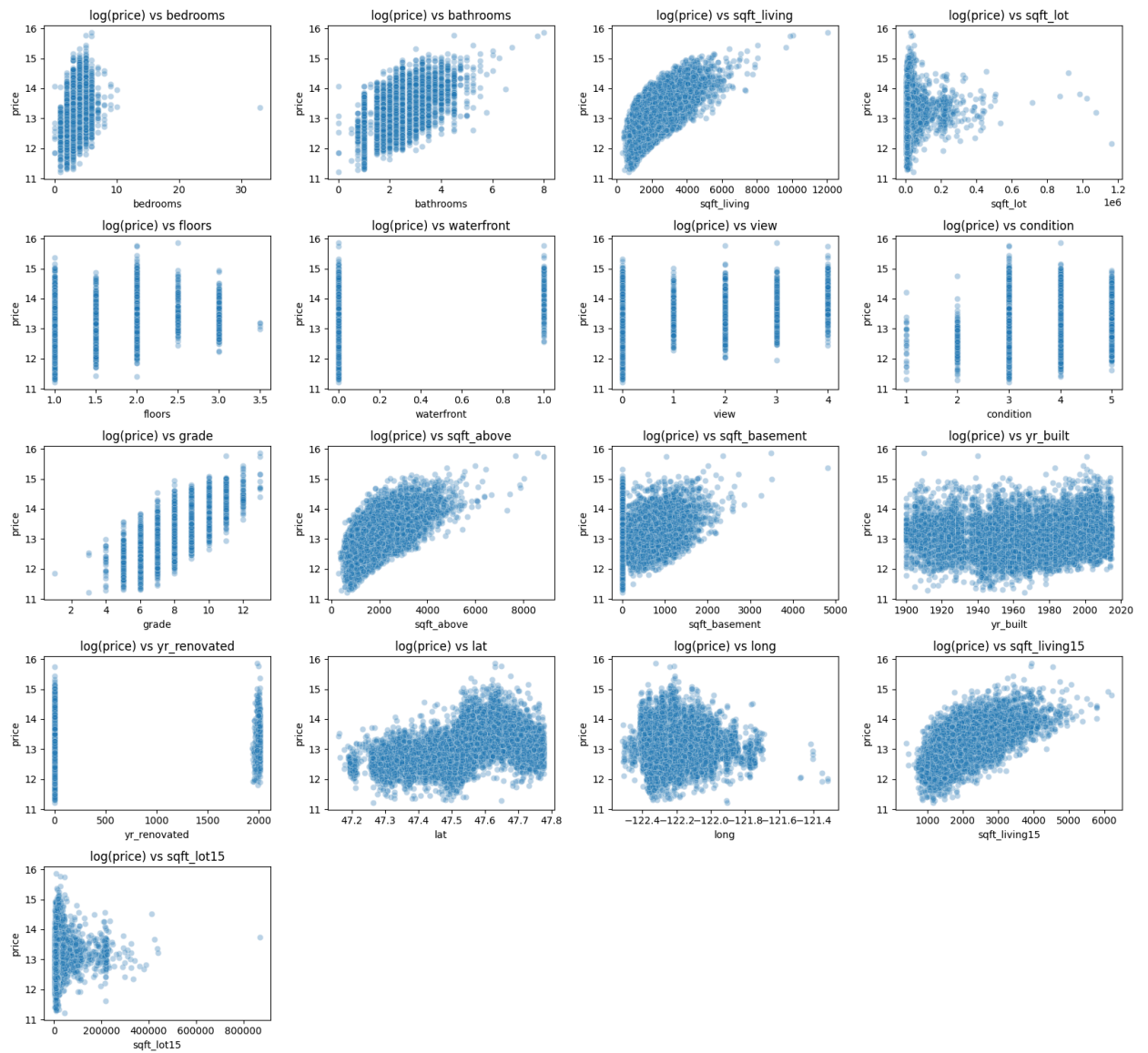
EDA

DATASET	SAMPLES	DESCRIPTION
Train	16209	18 tabular features + satellite image per property
Test	5404	18 tabular features + satellite image per property
Val	15%(train)	Used for model evaluation

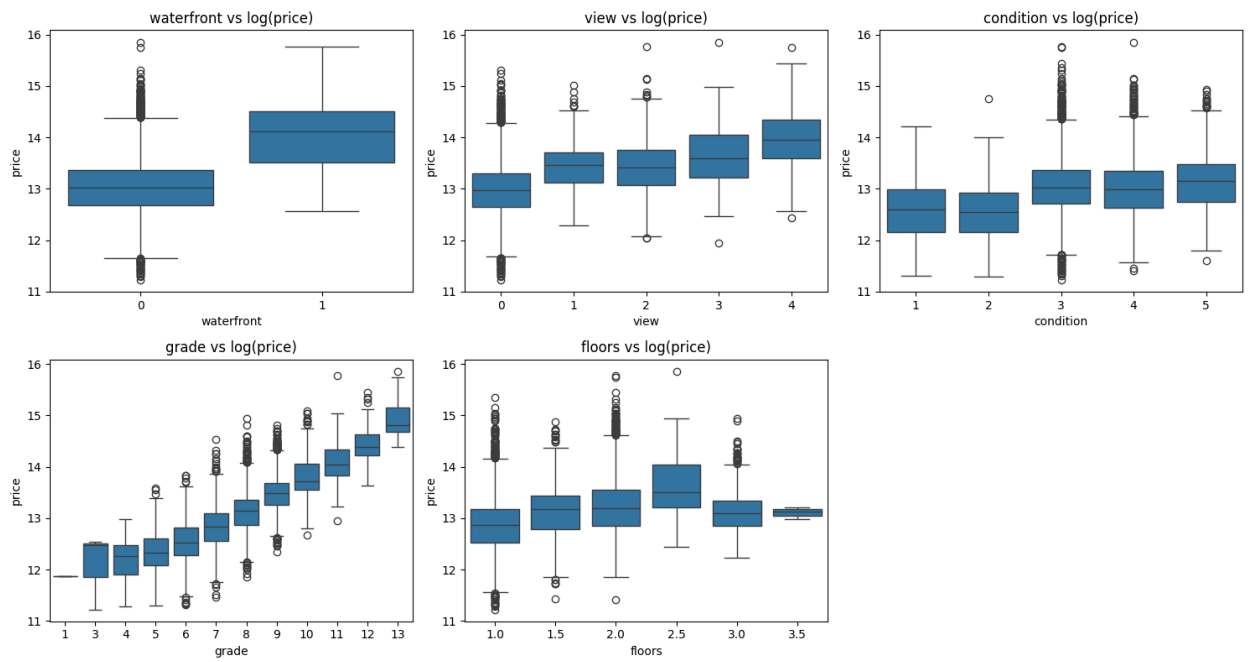
- Price-distribution:



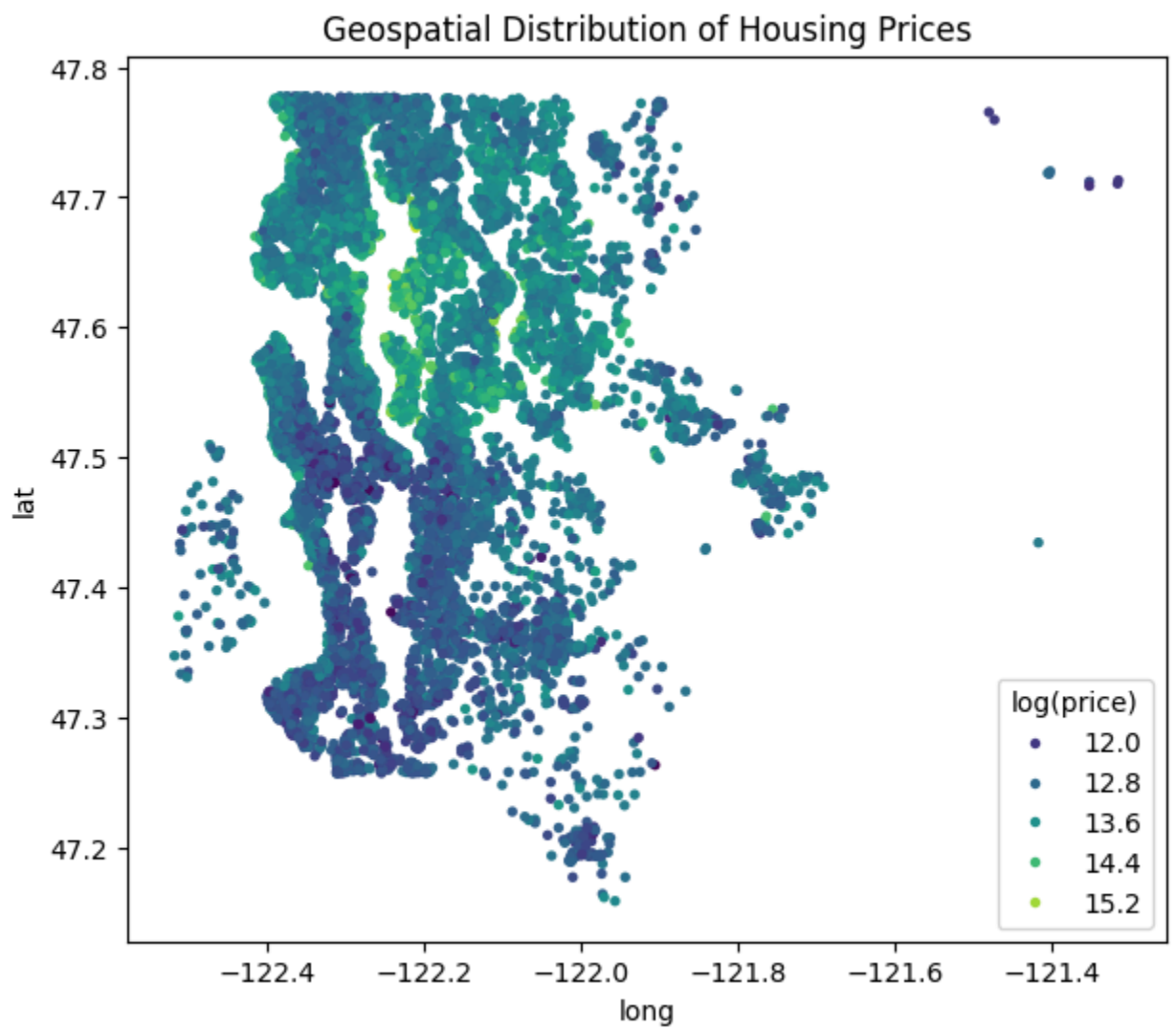
- Scatter-Plot:



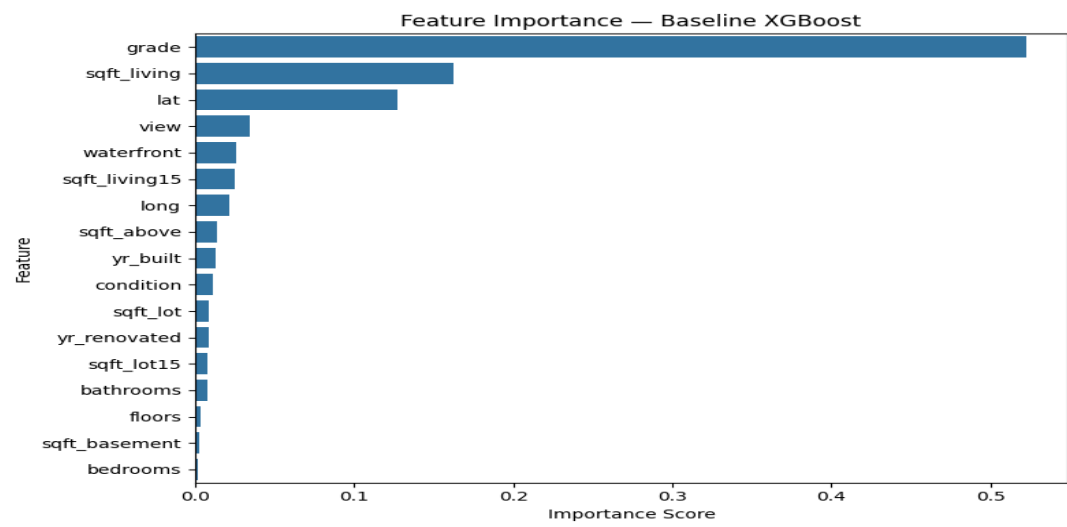
- Box-plot:



- House Distribution:

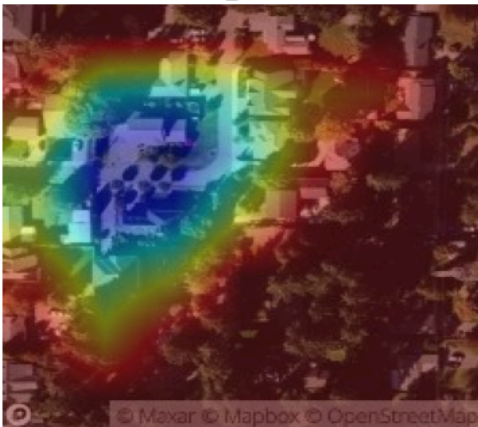


- Feature-Importance:

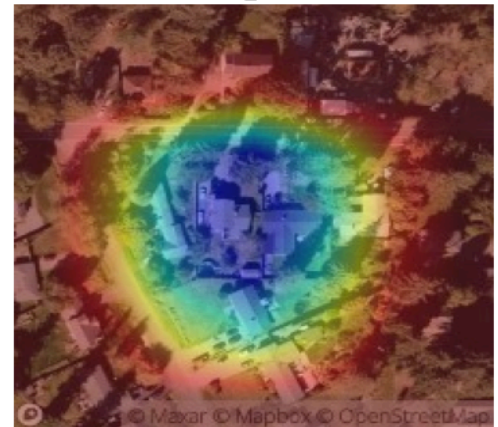


- Grad-cam:

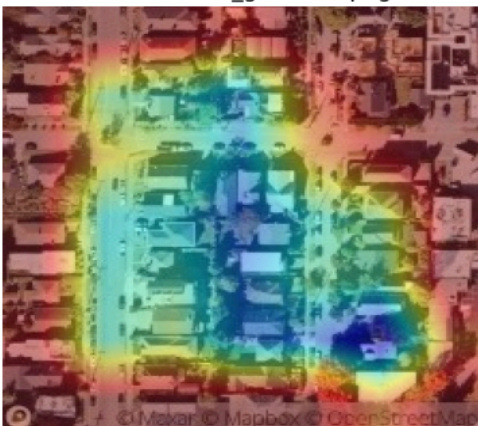
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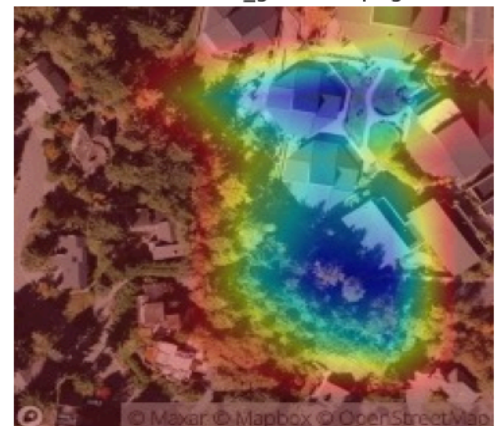
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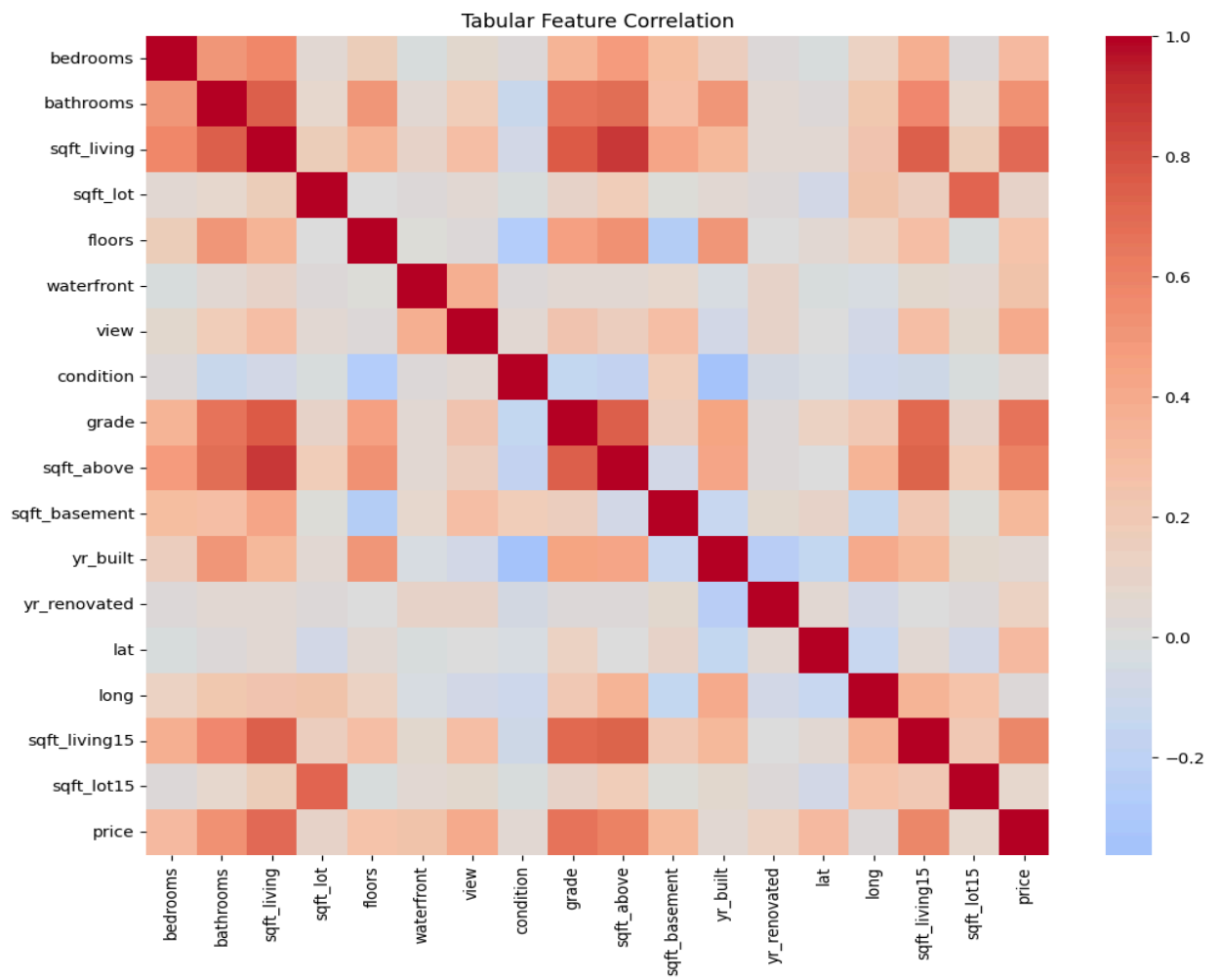
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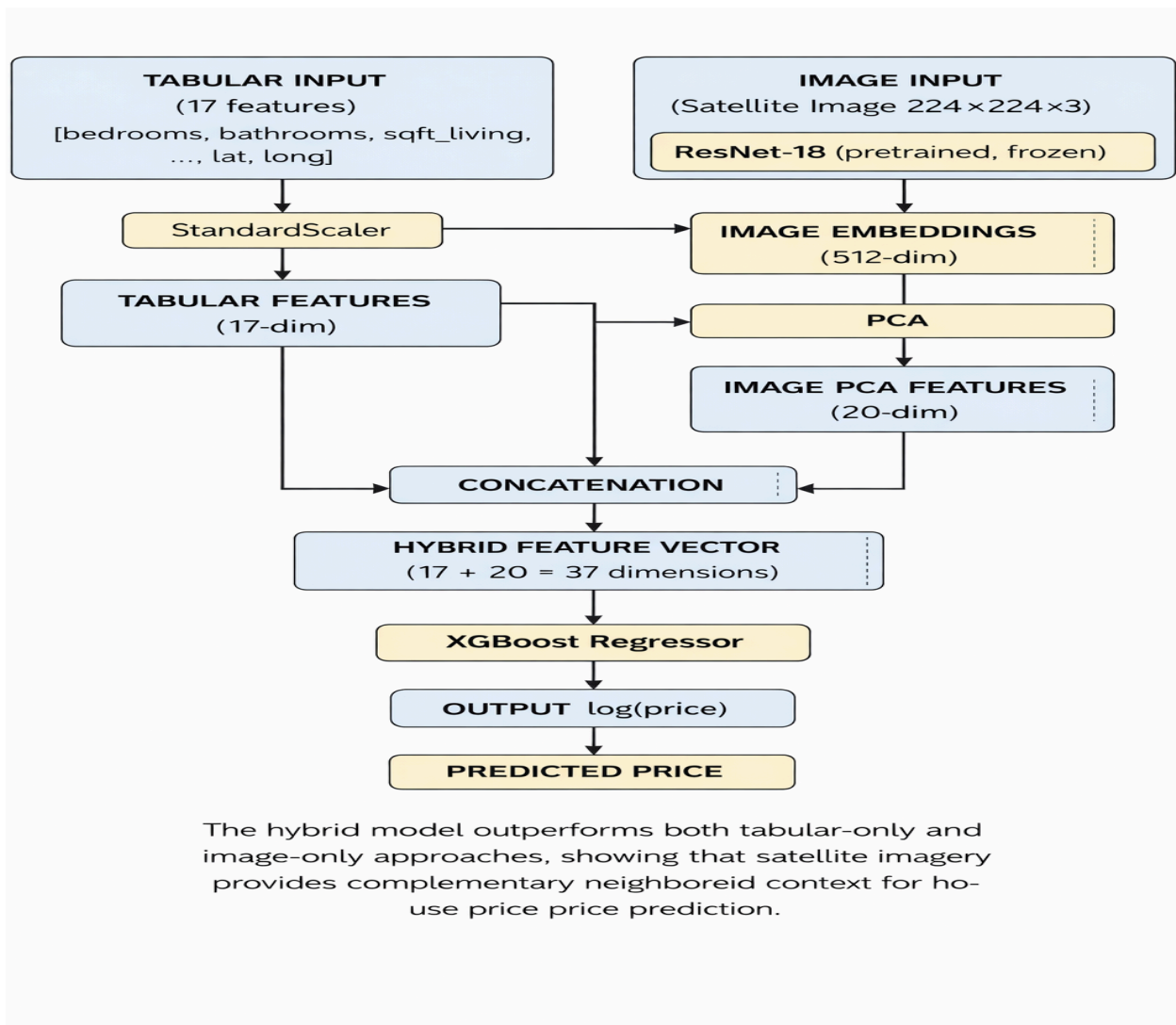
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- Grad-CAM highlights regions around **building footprints, road networks, and surrounding neighborhood structures**, indicating that the CNN captures spatial and environmental context.
- The model consistently focuses on **property-centered regions** rather than irrelevant background, suggesting meaningful visual feature extraction.
- Attention to nearby infrastructure and layout supports the contribution of satellite imagery in improving the hybrid model's price predictions.
- Feature Correlation:



ARCHITECTURE



CONCLUSION

The hybrid model combining tabular data and satellite imagery performs better than both tabular-only and image-only models. This demonstrates that visual neighborhood information provides complementary value for property price prediction beyond traditional features.

